

Santhosh Naidu Puppala

Pittsburgh, PA | 571-919-7165 | santhoshnaidu403@gmail.com | [linkedin.com/in/santhoshnaidupuppala](https://www.linkedin.com/in/santhoshnaidupuppala) | santhoshpuppala94.github.io/agentica

PROFESSIONAL SUMMARY

Enterprise Software Engineer with 8+ years of experience and 2-3 years of focused hands-on GenAI engineering across deployed AI delivery, LangGraph/LangChain-style agent workflows, MCP servers/clients, RAG, Python/FastAPI services, JavaScript/TypeScript integrations, evaluation harnesses, guardrails, observability, and enterprise workflow automation. Experienced working with technical and business stakeholders to turn ambiguous problems into production-oriented AI systems.

- Deployed AI delivery: customer/problem discovery, technical blueprinting, POCs, architecture tradeoff discussions, LangGraph orchestration, MCP servers/clients, reusable agent skills, RAG pipelines, FastAPI APIs, enterprise API integrations, monitoring pipelines, evaluation harnesses, and production support handoff.
- Engineering foundation: Python, JavaScript/TypeScript, Java, REST APIs, FastAPI, SQL, Git, CI/CD, Docker, Kubernetes concepts, Terraform concepts, AWS, Azure, GCP concepts, OpenAI API, Claude/Anthropic concepts, embeddings, semantic retrieval, Databricks, and PySpark.
- Production readiness and reliability: deployment validation, environment configuration, logging, monitoring, fault tolerance, agent reliability checks, cost/latency tracking, prompt/version tracking, groundedness checks, output validation, audit trails, privacy/security controls, incident response, and post-release improvements.
- Domain experience in financial services, healthcare data operations, telecom PLM, enterprise data platforms, and production support workflows.
- Strong fit for Deployed Engineer, Forward Deployed Engineer, AI Engineer, AI Native Software Engineer, Agentic AI Engineer, GenAI Engineer, LLM Application Engineer, AI Platform Engineer, and Python AI Backend roles.

TECHNICAL SKILLS

AI / LLM Engineering: AI agents, multi-agent systems, RAG, LLM integration, OpenAI API, Claude/Anthropic concepts, commercial/open-source LLMs, prompt engineering, structured outputs, grounded responses, hallucination controls, LLMOps

Agentic AI / Orchestration: LangGraph, LangChain concepts, MCP servers/clients, FastMCP, agent skills, tool/function invocation, multi-agent workflows, planner/supervisor/executor/evaluator patterns, policy-based routing, memory, task routing, human approval, fallback handling, workflow state tracking

Cloud-Native Backend: Python, JavaScript/TypeScript, Java, FastAPI, REST APIs, microservices, serverless/event-driven patterns, Pydantic, API Router, authentication/authorization, SQLAlchemy, async API development, SQL, PostgreSQL, SQLite, JSON contracts

Full Stack / Integration: React, TypeScript, JavaScript, API-driven UI integration, chat assistant interfaces, document Q&A screens, citation displays, feedback capture, enterprise API integration, dashboard components, stakeholder demo workflows

Data / Retrieval / Pipelines: embeddings, semantic search, hybrid retrieval, vector search, metadata filtering, re-ranking, ChromaDB, Pinecone, OpenSearch, GraphRAG concepts, Pandas, NumPy, Databricks, PySpark, data processing pipelines

Cloud / DevOps / Observability: AWS, Azure, GCP concepts, Docker, Kubernetes, Terraform concepts, CI/CD, GitHub, Jenkins, CloudWatch, Prometheus, Grafana, ELK, tracing, telemetry, latency/cost monitoring, monitoring pipelines

Evaluation / Governance: evaluation harnesses, agent reliability checks, response quality review, confidence scoring, groundedness checks, guardrails, safety controls, audit logging, Responsible AI, security, privacy, compliance, technical documentation

PROFESSIONAL EXPERIENCE

Senior AI/GenAI Engineer | PNC Bank (via DVI Technologies Inc.), Pittsburgh, PA | Oct 2024 - Present

- Built deployed-AI-style GenAI automation workflows using Python, LangGraph framework, LLM APIs, RAG, FastAPI, cloud-native services, and observability patterns to move from operational problem discovery to working AI prototypes and production-ready workflow automation.
- Defined technical blueprints and implementation tradeoffs for AI workflows spanning orchestration, retrieval/grounding, prompt and agent logic, MCP tool integration, evaluation, guardrails, serving, and observability.
- Developed reusable agent skills and LangGraph-based workflows with planner, supervisor, executor, evaluator, checkpointing, runtime context, memory, and workflow state tracking to automate multi-step production-support tasks.
- Implemented and tested MCP servers, MCP clients, custom tools, and MCP Inspector flows to connect AI agents with SQL databases, REST APIs, NAS repositories, logs, job-status lookups, and configuration systems.
- Designed RAG pipelines over enterprise runbooks, Confluence pages, SOPs, and operational documents using chunking, embeddings, vector search, metadata filtering, re-ranking, and citation-grounded retrieval to improve answer relevance.
- Built evaluation harnesses and guardrail patterns using expected-output checks, groundedness validation, confidence scoring, fallback escalation, human approval, audit trails, response-quality review, and failure-mode testing for reliable AI behavior.
- Developed FastAPI-based AI services and backend APIs using Python REST APIs, Pydantic validation, authentication, SQLAlchemy, async request handling, secure integrations, structured JSON contracts, and React/TypeScript UI integrations for stakeholder demos.
- Partnered with business, product, architecture, data, and engineering stakeholders to co-define problems, demo prototypes, align technical direction, explain architecture tradeoffs, refine AI outputs, and support production adoption.

GenAI Engineer / AI Data Engineer | SysArch Inc. / Zimmer Biomet, McLean, VA | Jun 2024 - Oct 2024

- Built a GenAI-assisted knowledge platform using Python, LLM/RAG patterns, vector retrieval, and data processing workflows to retrieve pipeline documentation, classify ETL failures, and generate structured recommendations for healthcare data teams.
- Developed an AI Data Quality Assistant that reviewed validation failures, schema mismatches, reconciliation issues, and pipeline anomalies, then produced actionable summaries for data operations teams.
- Designed enterprise ETL pipelines using Azure Data Factory, Databricks, and PySpark with metadata-driven transformations and data-quality validation for healthcare analytics workloads.
- Implemented LLM-powered semantic search across data dictionaries, schema definitions, and pipeline documentation to reduce manual investigation time for engineering teams.
- Built automated reconciliation workflows using PySpark and Databricks to validate source-to-target counts, null checks, referential integrity, and downstream reporting readiness.
- Created structured prompt templates for ETL error reports, root-cause summaries, impacted table lists, and remediation steps while respecting healthcare data governance constraints.

Sr. Software Engineer | Qentelli, Hyderabad, India | Jan 2022 - Jan 2023

- Designed and delivered production cloud-native enterprise applications on AWS using ECS, Elastic Beanstalk, CodeBuild, CodePipeline, Docker, and Kubernetes, improving deployment repeatability across environments.
- Built and maintained CI/CD pipelines with AWS CodeBuild and CodePipeline for Java and Python applications across development, staging, and production environments.
- Provisioned deployment infrastructure using Terraform including VPCs, subnets, IAM, S3, RDS, and security groups to support repeatable and secure production releases.

Senior Software Engineer | HCL Technologies (Client: Nokia), Bengaluru, India | Apr 2019 - Jan 2022

- Served as Informatica PowerCenter ETL middleware engineer for Nokia enterprise data platforms, designing mappings and workflows processing millions of records across 26+ downstream applications.
- Supported ENOVIA version upgrades by assessing downstream data impact, updating transformation logic, validating pipelines, and ensuring zero production data loss.
- Collaborated with business, data, and application teams to troubleshoot production issues, improve batch reliability, and maintain enterprise data flow quality.

Project Engineer | Wipro Limited (Client: Nokia), Bengaluru, India | Jun 2016 - Mar 2019

- Worked on Nokia ENOVIA PLM platform for BOM management, engineering change management, product configuration, and enterprise data workflows.
- Developed Java backend components and TCL/MQL automation scripts for PLM processes, upgrade validation, regression testing, and compatibility fixes.

PROJECTS

Enterprise GenAI Support Assistant | github.com/SanthoshPuppala94/enterprise-genai-support-assistant

- Built a production-style enterprise support assistant using the LangGraph framework for supervisor routing across Incident RCA, SQL, Document/RAG, Log Troubleshooting, and Letter Explanation agents.
- Implemented governed FastMCP tools/resources, MCP client boundaries, reusable agent skills, SELECT-only SQL safety, SQLite memory, citation-grounded RAG, incident runbook retrieval, code-change correlation, evaluation checks, human approval guardrails, and pytest validation.
- Stack: Python, FastAPI, LangGraph, FastMCP, SQLite, RAG, deterministic embeddings, MCP Inspector, pytest

MCP DeepAgent Workbench | github.com/SanthoshPuppala94/mcp-deepagent-workbench

- Built a production-style deep-agent workbench using LangGraph lifecycle control with FastMCP tools/resources, explicit state transitions, task decomposition, risk assessment, SQLite long-term memory, and deployable FastAPI execution flow.
- Implemented structured tool adapters, grounded-response checks, evidence-limited fallbacks, guardrail notes, FastAPI execution endpoint, and pytest coverage for routing, memory, MCP registration, and graph execution.
- Stack: Python, FastAPI, LangGraph, FastMCP, SQLite, StructuredTool patterns, agent state tracking, pytest

EDUCATION

Master's Degree | Wilmington University, New Castle, DE | May 2024 | GPA: 3.4